

☐ د٠☐ ه٠☐ اب١☐ اج١☐ اد١☐ اه١☐ إ٢☐ ب٢☐ ج٢☐ د٢☐ ه٢☐ إ٣☐ ب٣☒ ج٣☐ د٣☐ ه٣☐ إ٤☐ ب٤☐ ج٤☐ د٤☐ ه٤

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رقم الكمبيوتر *

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الكود * (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

LJLY

 **Will be reviewed**

8

How is a clipping plane defined in OpenGL? 

- ☐ By specifying four points that lie on the plane.
- ☒ By specifying the equation of the plane in the form $Ax + By + Cz + D = 0$.
- ☐ By specifying the normal vector to the plane and a point on the plane.
- ☐ By specifying the distance from the origin to the plane.

 **Will be reviewed**

9

When should glutMainLoop() typically be called in an OpenGL program? 

- ☐ At the beginning of the program.

- ☒ After initializing OpenGL and setting up the display callback function.
- ☐ After drawing the scene.
- ☐ Before creating the window.

 Will be reviewed

10

When is the timer callback function registered with glutTimerFunc invoked?



- ☐ Immediately after glutTimerFunc is called.
- ☒ After a specified interval, in milliseconds.
- ☐ When the window is resized.
- ☐ When a key is pressed.

 Will be reviewed

11

What is the difference between orthographic and perspective projection?



- ☒ Orthographic projection preserves parallel lines, while perspective projection does not.
- ☐ Perspective projection preserves parallel lines, while orthographic projection does not.
- ☐ Orthographic projection uses a single vanishing point, while perspective projection uses multiple vanishing points.
- ☐ Perspective projection uses a single vanishing point, while orthographic projection uses multiple vanishing points.

☑ Will be reviewed

23

How many parameters does glViewport take? 

☐ 2

☐ 3

☒ 4

☐ 5

☑ Will be reviewed

24

Write down the Modelview matrix after executing the following OpenGL code segment.

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);  
glScalef(-2.0, 1.0, -1.0);  
glRotatef(180.0, 0, 0, 1.0); 
```

☐ A

☒ B

☐ C

☐ D

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

☑ Will be reviewed

25

What happens inside the glutMainLoop() function? 

- ☐ It continuously checks for events like keyboard input, mouse input, and window resizing.
- ☐ It calls the registered display callback function to redraw the scene.
- ☐ It handles window management tasks like creating, destroying, and resizing windows.
- ☒ All of the above.

 **Will be reviewed**

26

The cross product of two vectors that are orthogonal is the zero vector. 

- ☐ True
- ☒ False The result is a vector that is perpendicular to both original vector

 **Will be reviewed**

27

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(180.0, 0, 0, 1.0);
glScalef(-1.0, 1.0, 1.0); 
```

- ☐ A
- ☐ B
- ☒ C

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

☐ D **Will be reviewed**

28

Can glViewport be used to create multiple viewports within a single window?

☐ Yes.☒ No. **Will be reviewed**

29

OpenGL supports z-buffering. 

☒ True☐ False **Will be reviewed**

30

Which matrix is used to project 3D objects onto a 2D screen? 

☐ Modelview matrix☒ Projection matrix☐ Texture matrix

☐ Viewport matrix

☑ Will be reviewed

31

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_STRIP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

☐ A

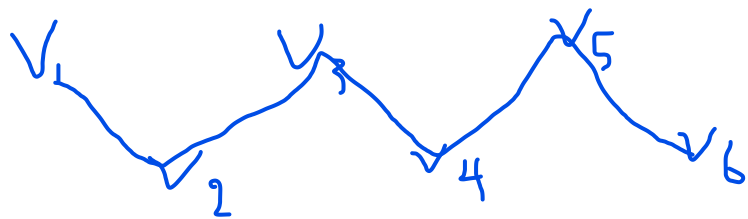
☐ B

☐ C

☒ D

☐ E

☐ F



☑ Will be reviewed

32

Can you set the alpha component of a color using glColor? 

☐ Yes.

`glColor4f(r, g, b, a);`

☒ No.

☑ Will be reviewed

33

What are the parameters typically passed to the glutReshapeFunc callback function? 

- ☒ The width and height of the window in pixels.
- ☐ The x and y coordinates of the mouse cursor.
- ☐ The current time in milliseconds.
- ☐ The state of the keyboard keys.

 **Will be reviewed**

34

In modern OpenGL, the user may supply shaders which will execute on the GPU. 

- ☒ True
- ☐ False

 **Will be reviewed**

35

When should glClear be called in a typical OpenGL rendering pipeline? 

- ☒ Before drawing any objects.
- ☐ After drawing all objects.
- ☐ Between drawing different objects.
- ☐ It doesn't matter when glClear is called.

✍ **Will be reviewed**

36

Order the following shader processes: 

1 **glCreateProgram**

1

2 **glAttachShader**

2

3 **glUseProgram**

4

4 **glLinkProgram**

3

✍ **Will be reviewed**

37

What is the primary function of glutTimerFunc in OpenGL? 

- ☐ To initialize the OpenGL environment.
- ☐ To create a window on the screen.
- ☐ To enter the main event processing loop.
- ☒ To register a timer callback function.

✍ **Will be reviewed**

38

When is the glutReshapeFunc callback function invoked? 

- ☐ When mouse is moved.

- ☒ When the window is resized.
- ☐ When a key is pressed.
- ☐ When the mouse is clicked.

 **Will be reviewed**

39

What is the primary function of **glRectf** in OpenGL? 

- ☐ Draws a filled rectangle.
- ☒ Draws a line.
- ☐ Draws a point.
- ☐ Clears the screen.

 **Will be reviewed**

40

What is the role of the glBegin and glEnd functions in relation to glVertex?



- ☒ They define the type of primitive to be drawn (e.g., points, lines, triangles).
- ☐ They specify the color of the vertices.
- ☐ They set the projection and modelview matrices.
- ☐ They control the depth testing and blending modes.

 **Will be reviewed**

41

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```

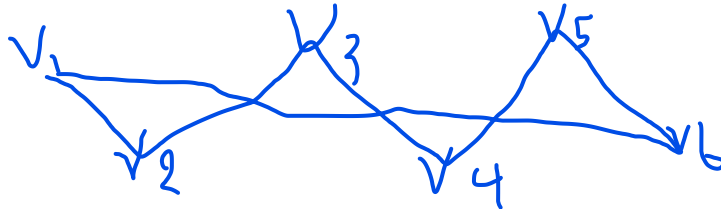
☐ A

☐ B

☒ C

☐ D

☐ E

☐ F


Will be reviewed

42

How does glColor affect the appearance of primitives drawn after it is called?


☐ It sets the color for all subsequent primitives until a new glColor call is made.

☒ It only affects the color of the next primitive.

☐ It affects the color of all primitives in the scene.

☐ It only affects the color of the preceding primitive.

Will be reviewed

43

What is the difference between GL_LINES and GL_LINE_STRIP? 

- ☒ GL_LINES draws individual lines, while GL_LINE_STRIP connects vertices with lines.
- ☐ GL_LINES connects vertices with lines, while GL_LINE_STRIP draws individual lines.
- ☐ GL_LINES draws a closed polygon, while GL_LINE_STRIP draws an open polygon.
- ☐ There is no difference between the two.

 **Will be reviewed**

44

How many parameters does **glRectf** take? 

- ☒ 2
- ☐ 4
- ☐ 6
- ☐ 8

 **Will be reviewed**

45

What is the relationship between glClear and glClearColor? 

- ☐ **glClearColor sets the color to clear the color buffer, while glClear actually clears the buffer.**
- ☐ glClear sets the color to clear the color buffer, while glClearColor actually clears the buffer.
- ☒ Both functions are used to clear the color buffer, but with different parameters.

☐ There is no direct relationship between glClear and glClearColor.

 **Will be reviewed**

46

Which type of projection is commonly used in video games and 3D simulations? 

- ☐ Orthographic projection
- ☒ Perspective projection
- ☐ Metric projection
- ☐ Oblique projection

 **Will be reviewed**

47

Which type of projection is commonly used for technical drawings and blueprints? 

- ☐ Orthographic projection
- ☐ Perspective projection
- ☐ Metric projection
- ☐ Oblique projection

 **Will be reviewed**

48

What do the parameters of glRectf represent? 

- ☐ The center point and radius of the rectangle.
- ☐ The top-left and bottom-right corners of the rectangle.
- ☒ The width and height of the rectangle.
- ☐ The color of the rectangle.

 **Will be reviewed**

49

The parameters of the glViewport function typically specify: 

- ☐ a) The width and height of the viewport
- ☐ b) The x and y coordinates of the viewport's lower-left corner
- ☒ c) Both (a) and (b)
- ☐ d) None of the above

 **Will be reviewed**

50

What is the primary function of glClipPlane? 

- ☒ To define a clipping plane in 3D space.
- ☐ To clip a polygon against a rectangular region.
- ☐ To remove hidden surfaces from a 3D scene.

- ☐ To apply texture mapping to polygons.

 **Will be reviewed**

51

What is the purpose of the `glPolygonMode` function? 

- ☐ To set the color of a polygon
- ☐ To set the texture of a polygon
- ☒ To set the mode in which polygons are rendered
- ☐ To set the depth test function

 **Will be reviewed**

52

What is the primary function of `glVertex` in OpenGL? 

- ☒ To define a vertex in 3D space.
- ☐ To draw a line between two points.
- ☐ To set the color of vertices.
- ☐ To set the color of subsequent vertices.

 **Will be reviewed**

53

Which of the following is NOT a valid mode for `glPolygonMode`? 

☐ GL_POINT☒ GL_LINE☐ GL_FILL☐ GL_WIREFRAME

 **Will be reviewed**

54

What is the difference between glVertex2f and glVertex3f? 

- ☒ glVertex2f defines 2D vertices, while glVertex3f defines 3D vertices.
- ☐ glVertex2f uses integer coordinates, while glVertex3f uses floating-point coordinates.
- ☐ There is no difference between the two functions.

 **Will be reviewed**

55

How many parameters does glutTimerFunc take? 

☐ 1☒ 2☐ 3☐ 4

 **Will be reviewed**

56

OpenGL Shading Language (GLSL) supports pointers. 

☒ True

☐ False

 **Will be reviewed**

57

When using the Phong Reflectance Model, we calculate the red, green, and blue color channels independently. 

☒ True

☐ False



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